

Customer Churn Analytics Report

Predictive Insights & Retention Strategy

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Executive Summary

This report presents a comprehensive analysis of customer churn within a telecommunications context, combining historical data, machine learning predictions, and interactive Power BI dashboards. The analysis identifies a churn rate of approximately 26.5%, with predictive outputs closely aligning to actual outcomes, indicating strong model performance and reliability.

- ~24% of customers fall into high or very high churn risk categories
- Month-to-month contracts represent the highest-risk segment
- Fiber optic customers exhibit elevated churn compared to other services
- Short tenure and higher monthly charges are primary churn drivers

The integration of predictive analytics enables a transition from reactive reporting to proactive retention strategy, allowing targeted intervention and improved customer lifetime value.

Introduction

Customer churn remains a critical business challenge, directly impacting revenue stability, acquisition costs, and long-term growth. This project integrates data analysis, machine learning, and business intelligence to identify churn drivers, quantify risk, and enable targeted retention strategies.

Hypothesis

- Customers with shorter tenure are more likely to churn
- Month-to-month contracts increase churn probability
- Higher monthly charges contribute to price sensitivity
- Fiber optic service users may have higher expectations and dissatisfaction risk

Key Business Questions

- What factors drive customer churn?
- Which customers are most at risk?
- How should retention efforts be prioritized?
- How can predictive analytics improve decision-making?

Data Overview

The dataset includes customer demographics, account information, service usage, billing metrics, and churn outcomes.

- Customer demographics (e.g., gender, partner status)
- Account details (contract type, tenure, charges)
- Service usage (internet type, add-ons)
- Churn outcomes and ML predictions

Dashboard Insights

Executive Overview

- Total Customers: 7,043
- Actual Churn: 1,869 (~26.5%)
- Predicted Churn closely aligned with actual outcomes
- High-risk customers: 1,688

Contract type and service usage are the strongest drivers of churn.

Retention Action

- 901 very high-risk customers identified
- High-risk customers concentrated in month-to-month contracts
- Short tenure and high charges dominate top-risk segments

This enables targeted intervention and prioritised retention efforts.

ML Model Insights

- Top drivers: Total Charges, Tenure, Monthly Charges
- Service type and payment method influence churn
- Model outputs align with expected business behaviour

Conclusion

The analysis confirms that churn is primarily driven by contract flexibility, tenure, pricing, and service type. The machine learning model provides reliable, interpretable predictions that enable proactive customer retention strategies.

Recommendations

Immediate Actions

- Target very high-risk customers with personalised retention offers
- Prioritise urgent intervention cases identified in the model outputs

Strategic Initiatives

- Encourage transition to long-term contracts
- Investigate fiber optic service dissatisfaction
- Review pricing strategy for high-charge customers

Future Enhancements

- Integrate customer lifetime value (CLV)
- Implement automated retention recommendations
- Develop real-time prediction capabilities